

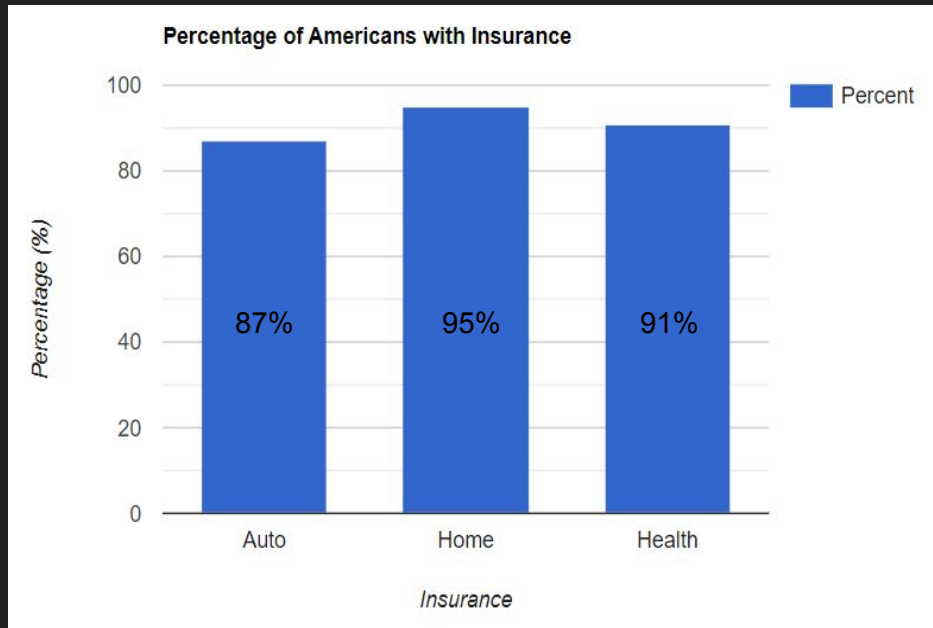
My SMART Program Project - Insurance: The Safety Net of Society

By: Cameron
Fong



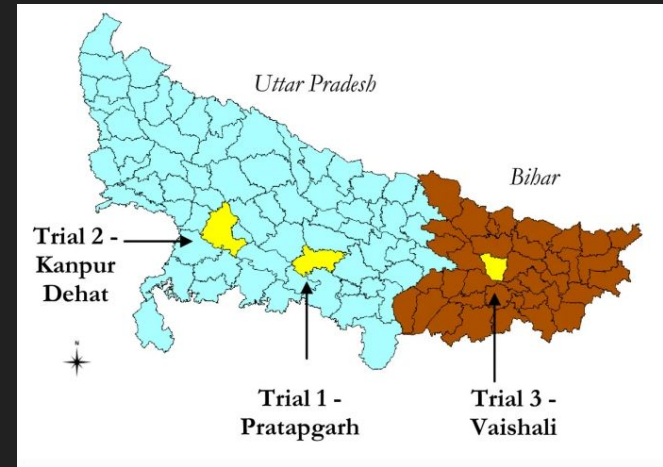
Context

- Insurance: Policy between company and customers
 - Company's contribution: money for damage, loss, etc.
 - Customer's contribution: monthly or yearly premium, also known as insurance premium
- Insurance is part of nearly every American's life
 - Almost every category is above 90%
- Insurance significantly varies by person
 - Factors include:
 - Age
 - Gender
 - Children



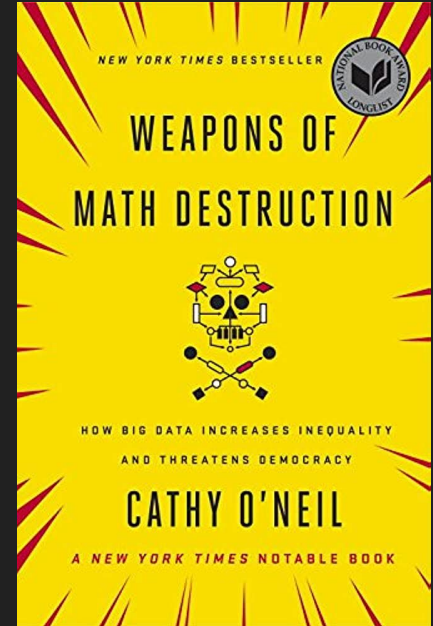
Articles that Motivated Me: Micro Health Insurance Project

- Test effectiveness of community based health insurance to combat steep price of insurance
 - Branch of Micro Health Insurance
 - Used 23,876 people from Uttar Pradesh India
 - Split into 3 clusters of testing
- Methods used
 - Randomization of villages - control bias
 - Quantitative data - household surveys for health status
 - Qualitative data - types of health issues
- Impact on research
 - Gave ideas to split data into regions
 - Helped give inspiration to use insurance as main subject



Book that Motivated Me: Weapons of Math Destruction

- Insurance grew out of actuarial science
 - How insurance is calculated: Factors
 - Includes credit score, safety of environment, etc.
 - Issue: Insurance premium based on background rather than actual performance (notably an issue with auto insurance)
 - Both background and actual performance face the same issue: privacy
- Impact to project
 - Primary factor to choosing research topic of interest
 - Inspiration for finding patterns within data
 - Specifically how much impact each factor of a person has on topic of interest



The Question

How does a person's background (gender, age, region, number of children, etc.) impact a person's insurance premium?

Why did I choose to pursue this subject?

- Curious how much daily life can play into risk factors that companies calculate to ultimately come up with an insurance plan

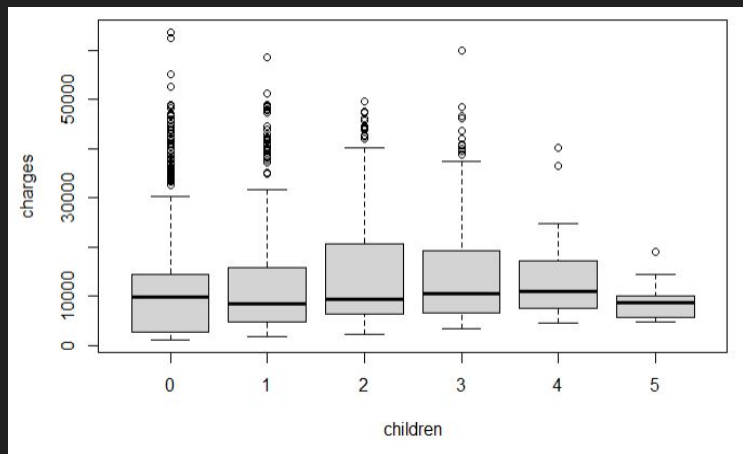
Database

- 1337 randomized people from America
 - Ages: 18-64
 - Factors: age, sex, bmi, children, smoker, region
 - Result: charges per year (insurance premium)

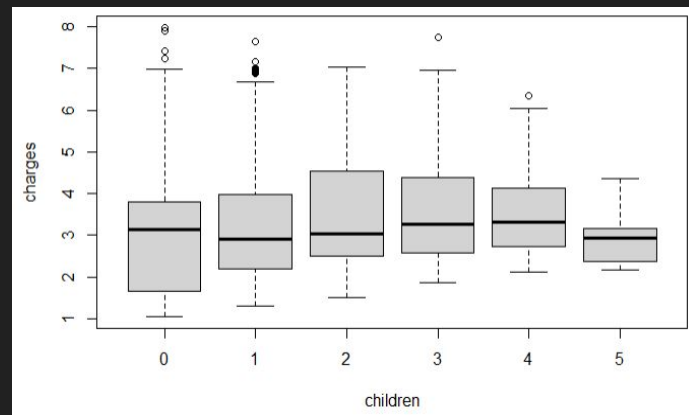
age <dbl>	sex <chr>	bmi <dbl>	children <dbl>	smoker <chr>	region <chr>	charges <dbl>
19	female	27.900	0	yes	southwest	16884.924
18	male	33.770	1	no	southeast	1725.552
28	male	33.000	3	no	southeast	4449.462
33	male	22.705	0	no	northwest	21984.471
32	male	28.880	0	no	northwest	3866.855
31	female	25.740	0	no	southeast	3756.622
46	female	33.440	1	no	southeast	8240.590
37	female	27.740	3	no	northwest	7281.506
37	male	29.830	2	no	northeast	6406.411
60	female	25.840	0	no	northwest	28923.137

Issue #1: Hard to Read Information

- Charges would be in the tens of thousands
 - Actual values for box plots and charts are difficult to read
- Solution: reduce charges by 1000, table for values
 - Charges are also square root transformed for issue #2



Before



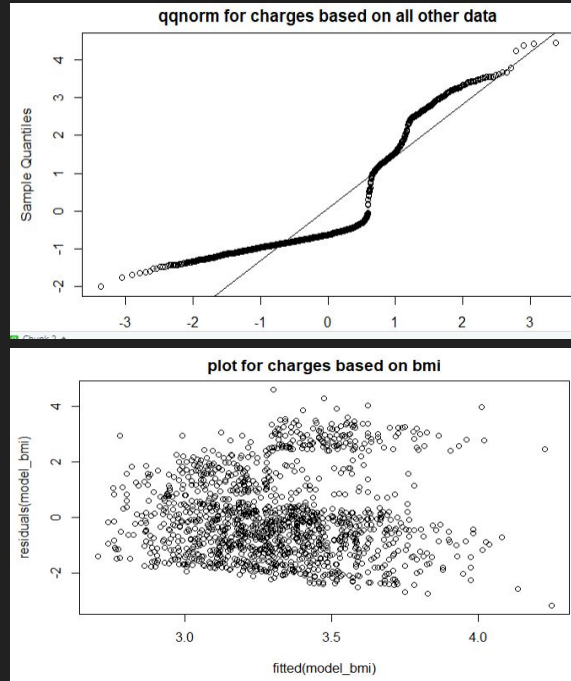
After

	1 child	2 children	3 children	4 children	5 children	6 children
Min	1.059185	1.308062	1.517894	1.855550	2.122419	2.165132
First Quartile	1.652850	2.188064	2.501562	2.579250	2.740852	2.369677
Median	3.139578	2.912693	3.043836	3.255848	3.321696	2.930796
Third Quartile	3.801296	3.987520	4.536398	4.381774	4.138650	3.177573
Maximum	6.991791	6.670937	7.041141	6.967724	6.048164	4.361566

Issue #2: Splitting Data

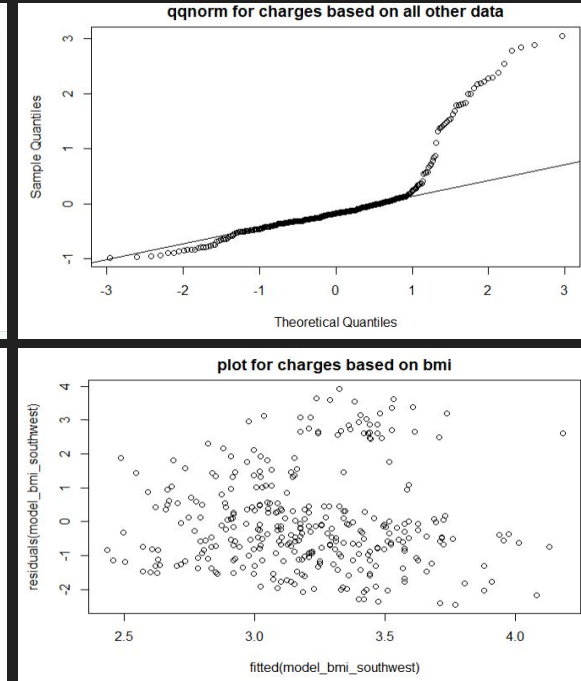
- Data for QQplot isn't normal
 - Residuals have unknown gaps
- Solution: Separating data points into 4 groups
 - Northwest, Northeast, Southwest, Southeast
 - Allows for elimination of residual patterns
 - Plots follow more closely to linear model

Before



Data from all regions

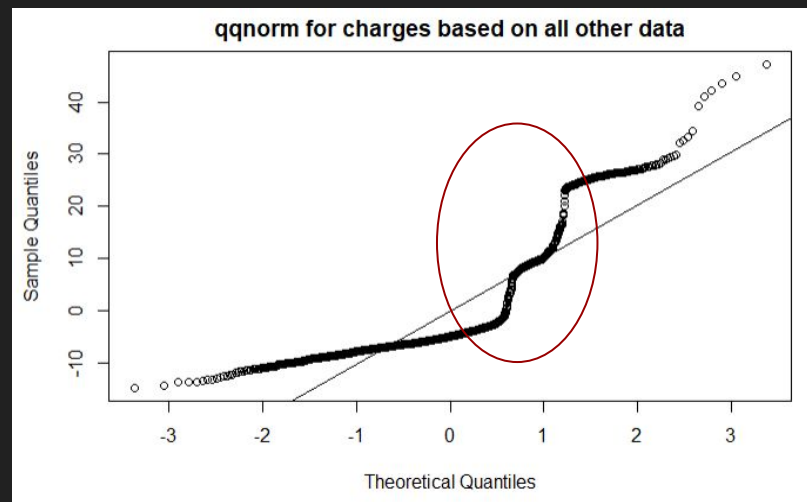
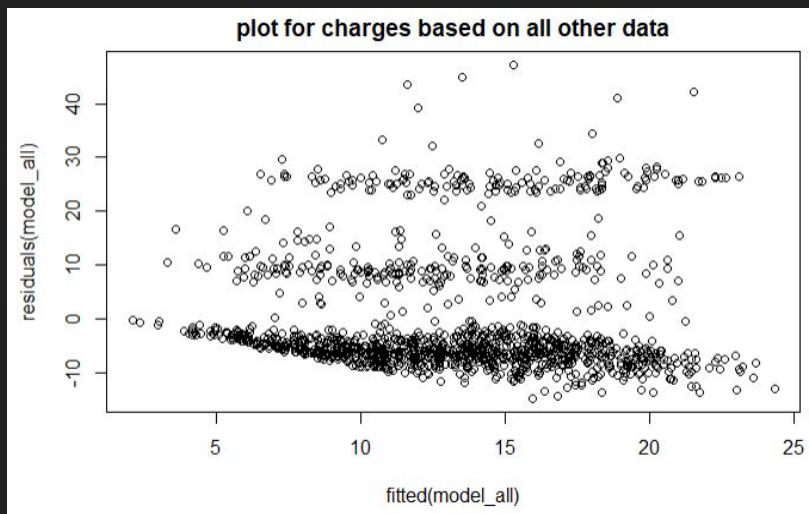
After



Data from Southwest region

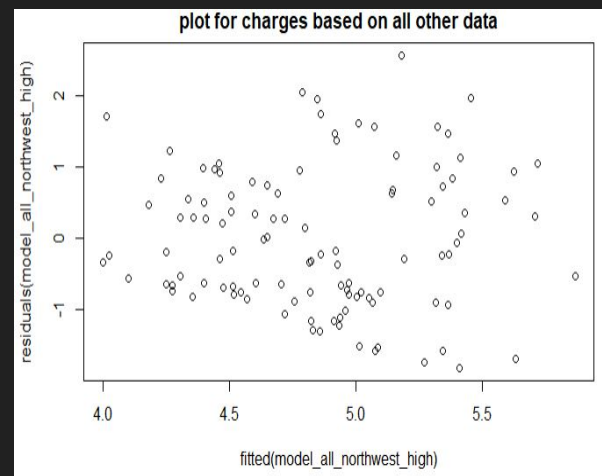
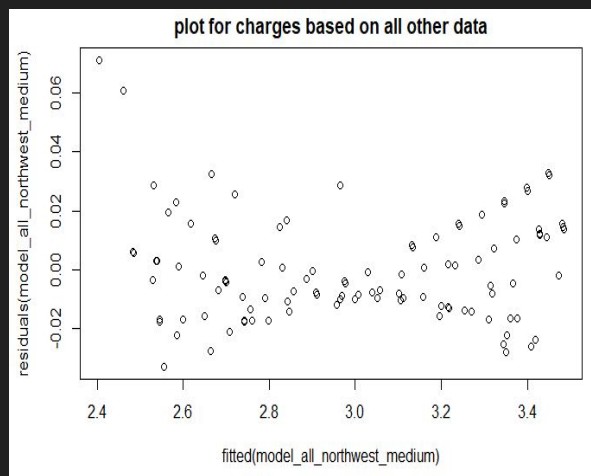
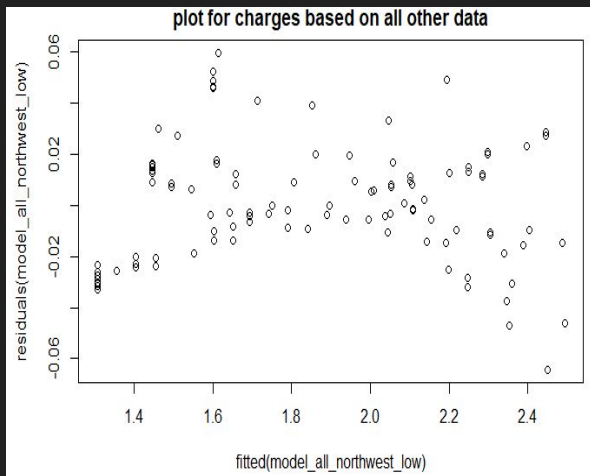
Issue #3: 3 Sections of Residuals

- QQ plots and residual charts would show 3 clusters of plots rather than following linear model and randomized data points respectively

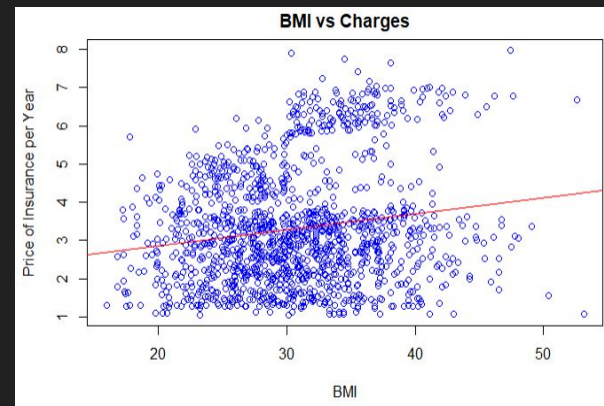
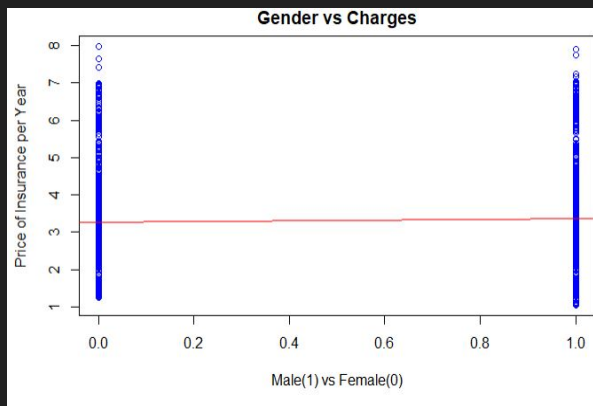
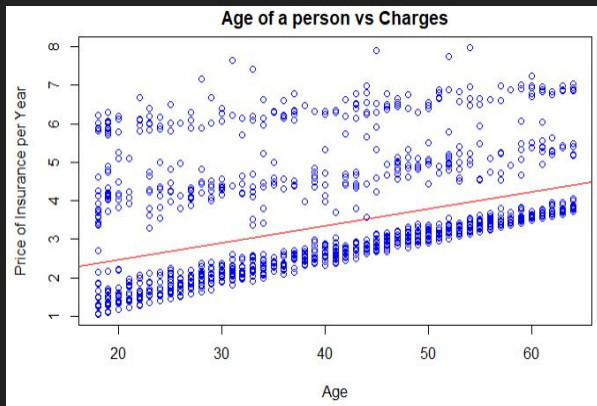


Issue #3: 3 Sections of Residuals (cont.)

- Solution: Split each region into 3 sections of premium: lowest 33%, middle 33%, upper 33%
- Reasoning: Ensures consistency in data models among the overall samples



Exploratory Data Analysis Results

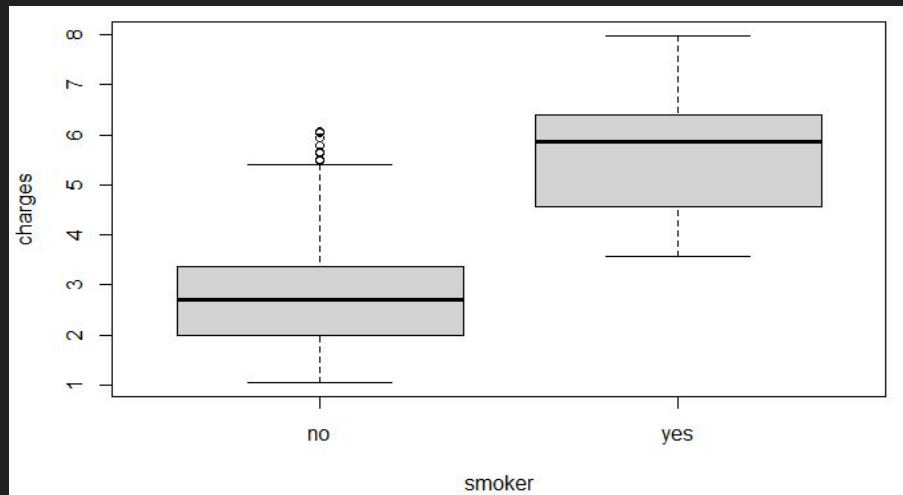


Intercept	1.584 (p-value < 0.05)
Age	0.044 (p-value < 0.05)
Interpretation	Approximately \$1.93 per age increase

Intercept	3.259 (p-value < 0.05)
Gender	0.111 (p-value > 0.05)
Interpretation	Approximately \$12.43 increase for being male

Intercept	2.041 (p-value < 0.05)
BMI	0.042 (p-value < 0.05)
Interpretation	Approximately \$1.72 per BMI increase

Exploratory Data Analysis (cont.)

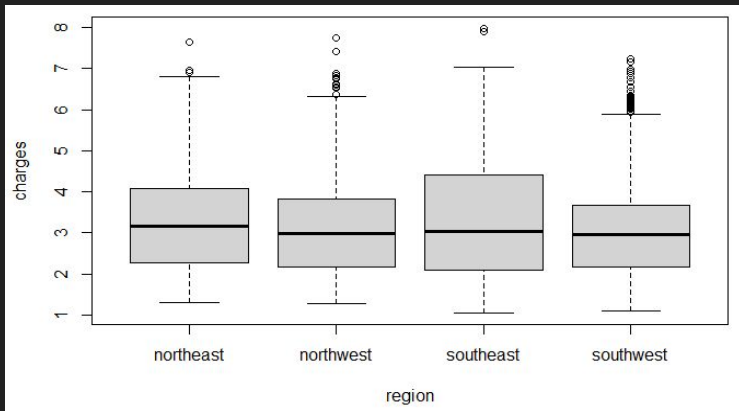


- Smoking significantly increases charges
- On average, 27111.62 more

	no	yes
Min	1.059185	3.581823
First quartile	1.996234	4.557810
Median	2.710241	5.869953
Third quartile	3.370908	6.405796
Maximum	5.402452	7.985639

Smoking	No	Yes
Min	1121.87	12829.47
1st quartile	3984.95	20773.63
Median	7345.41	34456.35
3rd quartile	11363.02	41034.22
Maximum	29186.49	63770.43

Exploratory Data Analysis (cont.)

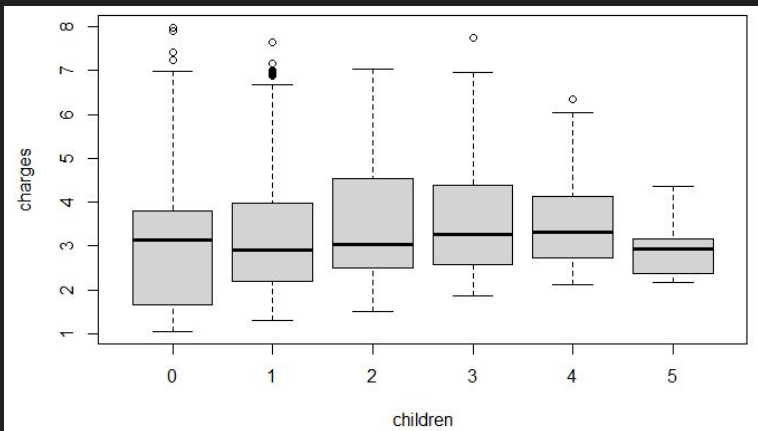


	northeast	northwest	southeast	southwest
Min	1.301843	1.273319	1.059185	1.114255
First Quartile	2.275746	2.172495	2.105301	2.179695
Median	3.171379	2.994294	3.048627	2.966242
Third quartile	4.088638	3.835589	4.419344	3.669131
Maximum	6.801111	6.324819	7.041141	5.902446

- Conclusion: all regions have similar average charge cost

Region	Northeast	Northwest	Southeast	Southwest
Min	1694.80	1621.34	1121.87	1241.56
1st quartile	5179.02	4719.73	4432.29	4751.07
Median	10057.64	8965.80	9294.13	8798.59
3rd quartile	16716.96	14711.74	19530.60	13462.52
Maximum	46255.11	40003.33	49577.67	34838.87

Exploratory Data Analysis (cont.)



	0 children	1 children	2 children	3 children	4 children	5 children
Min	1.059185	1.308062	1.517894	1.855550	2.122419	2.165132
First Quartile	1.652850	2.188064	2.501562	2.579250	2.740852	2.369677
Median	3.139578	2.912693	3.043836	3.255848	3.321696	2.930796
Third Quartile	3.801296	3.987520	4.536398	4.381774	4.138650	3.177573
Maximum	6.991791	6.670937	7.041141	6.967724	6.048164	4.361566

- Conclusion: the number of children relatively keeps charges consistent

Children	0 children	1 child	2 children	3 children	4 children	5 children
Min	1121.87	1711.03	2304.00	3443.07	4505.66	4697.80
1st quartile	2731.91	4787.62	6257.81	6652.53	7512.27	5615.70
median	9856.95	8483.78	9264.94	10600.55	11033.66	8589.57
3rd quartile	14449.85	15900.31	20578.90	19199.94	17128.42	10096.97
maximum	48885.14	44501.40	49577.67	48549.18	36580.29	19023.26

Multiple Linear Regression Results

	Intercept	Age	Gender	Bmi	Children	Smoking
All Southwest	-0.056	0.047	n/a	0.025	0.048	3.086
Southwest lowest 33%	0.314	0.052	-0.145	n/a	0.157	n/a
Southwest middle 33%	1.143	0.037	-0.05	n/a	0.057	n/a
Southwest upper 33%	1.11	0.008	n/a	0.076	n/a	1.82
All southeast	-0.218	0.046	n/a	0.029	0.111	3.143
Southeast lowest 33%	0.313	0.052	-0.15	n/a	0.162	n/a
Southeast middle 33%	1.329	0.034	n/a	n/a	0.043	n/a
Southeast upper 33%	0.714	0.017	n/a	0.093	n/a	1.319

	Intercept	Age	Gender	Bmi	Children	Smoking
All Northwest	0.056	0.042	n/a	0.029	0.142	2.613
Northwest lowest 33%	0.509	0.049	-0.1396	n/a	0.154	n/a
Northwest middle 33%	0.42	0.052	-0.083	n/a	0.097	n/a
Northwest upper 33%	1.5936	n/a	n/a	0.087	n/a	1.254
All Northeast	-0.171	0.04	n/a	0.044	0.126	2.561
Northeast lowest 33%	0.573	0.047	-0.131	0.001	0.146	n/a
Northeast middle 33%	1.921	0.021	n/a	0.006	n/a	1.157
Northeast upper 33%	1.325	n/a	n/a	0.098	n/a	1.213

Multiple Linear Regression - Conclusions

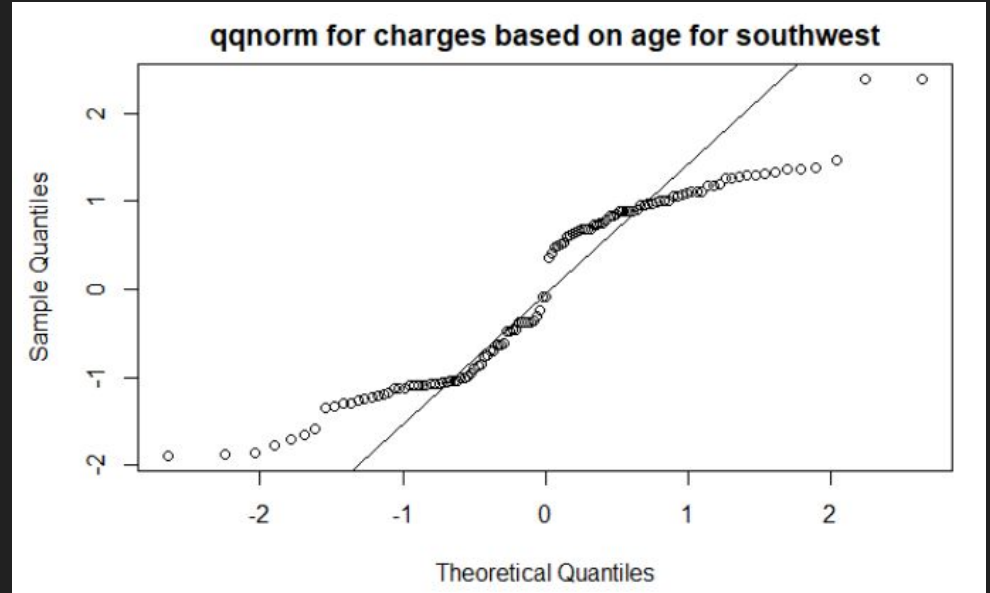
- Children increase charges significantly for lower charges
- Smoking increases significantly the price of insurance
- BMI/smoking largest typically only factors for high charges
- BMI almost never a factor for small and medium charges
- Age relatively same for all charges, but slightly decreases as a factor when insurance prices increase
- Being female noticeably shown for lower charges

Closing Thoughts - What Did I Learn?

- Importance of normality and constant variances
- Extracting and cleaning data
- Creating easy to read infographics and charts
- Analyzing why patterns occur within the data

Closing Thoughts - Limitations

- Time to figure why some data had unusual skews
 - Likely due to misplacement of a few subjects in incorrect premium group
- More information necessary about individuals
 - Effects outside of the data not shown within charts
 - Ex: Income
- More individuals for sample
 - More accurate results



Closing Thoughts - Future Work

- Larger sample size for more accurate results
- Sample size based on specific population
 - Ex: population of Riverside
- Answering different questions on the subject of insurance
 - Ex: factors customers look for when choosing a specific insurance plan

Thank you!